



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

ICS 2203 WEB APPLICATION DEVELOPMENT 1

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 8

XML

Learning outcomes

Upon completing this topic, you should be able to:

1. Identify XML web technologies used in carrying data.
2. Employ XML technology carrying web data.

8.1. Introduction

XML stands for extensible Markup Language. XML is a language that was designed to transport and store data. It is a markup language much like HTML that was designed too carry and transport Data. Unlike HTML, XML tags are not prede-



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fined. You must define your own tags. XML is designed to be self-descriptive XML has been also instrumental in creating new internet languages such as XHTML, WSDL(web service description language), RSS(really simple syndicate) for news feeds e.t.c XML is able to carry data through the syntax that may be used to represent semi structured data.

To take you slightly back in the theory of databases you may remember learning about data bases and data models. There are a number of data models in use for different applications but the most popular is the relational model. The relational model defined data in form of tables or relations that may be linked. Other data models include the Object oriented data model and theoretically in this model data is stored in objects and only accessed through set interfaces. For the purposes of



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this lesson we have the Semi structured model which views data in the form of a tree structure. This model is renowned for its flexibility in presentation of data and allowing for sharing of the same. This makes it ideal for use on the web when applications are to share data across the internet. XML allows application to application sharing of data.

8.1.1. What is a semi Structured data Model

It's basically a tree structure with an arbitrary number of children for each node. A semi-structured data model is optimal for one kind of data only: semi-structured data. Data is organized in semantic chunks (entities), similar entities are grouped together (relations or classes) entities in the same group have the same descriptions (attributes). Descriptions for all entities in a



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group (schema) have the same defined format have a predefined length are all present and follow the same order

8.1.2. Why XML?

XML presents a natural way of working with semi structured data. It is capable of carrying data represented in this type of a data model. in addition to this xml is known to have the following:

- Provides a high degree of interoperability with other applications
- Has no ties to any platform or language. It is an open standard
- It can be tailored to any users specifications



- With its clear, simple syntax and unambiguous structure, XML is easy to read and parse by humans and programs alike.
- XML is easily combined with style sheets to create formatted documents in any style you want. The purity of the information structure does not get in the way of format conversions.

8.1.3. Graphs and Semi structured data

- Nodes = objects.
- Labels on arcs (like attribute names).
- Atomic values at leaf nodes (nodes with no arcs out).
- Flexibility: no restriction on:



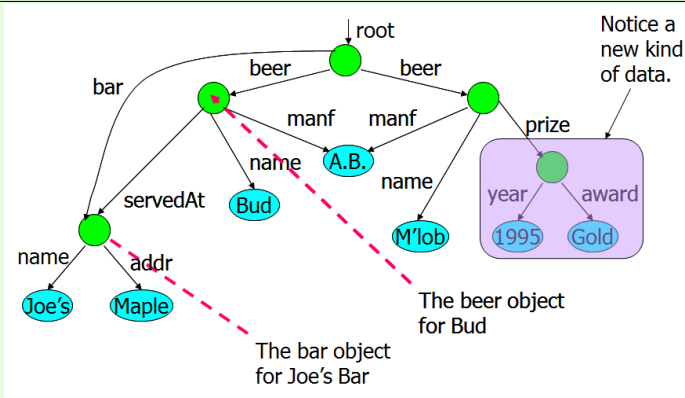
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- Labels out of a node.
- Number of successors with a given label

Example. Example of Graph

The Graph below represents data being carried. The actual data is what is in the leafs that is Joes, Maple, Bud, A.B, M'lob, 1995 and gold.

document additionally conforms to semantic rules,



Well-Formed and Valid XML

A Well-Formed XML allows you to invent your own tags. Valid XML conforms to a certain DTD. XML document start with a declaration, surrounded by `<?xml ... ?>`.

Normal declaration is:



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```
<?xml version = "1.0" standalone = "yes" ?>
```

“standalone” = “no DTD provided.”

Balance of document is a root tag surrounding nested tags.

Tags are normally matched pairs, as <FOO> ... </FOO>.,

Unmatched tags also allowed, as <FOO/> Tags may be nested arbitrarily and are case sensitive

Below is a representation of our previous graph in XML

Notice the labels automatically form the user defined tags while the leaf nodes are sandwiched between these tags. Notice also the nesting of these tags in the form of searching a tree starting from the root to the branches

```
?<? xml version = "1.0" standalone = "yes"?>
```

```
<BEER><NAME>Bud</NAME> <MANF>A.B</MANF>
```

```
<SERVEDAT><NAME>JOES BAR</NAME>
```



```
<ADDR></ADDR>MAPLE</SERVEDAT>  
</BEER> <BEER>><NAME>Bud</NAME>  
<MANF>A.B</MANF> <PRIZE><YEAR>1995</YEAR>  
<AWARD>GOLD</AWARD></PRIZE> </BEER> ...  
</BARS>
```

8.1.4. Document Type Definition (DTD)

A DTD specifies the syntax of a web page in SGML Standard Generalized Markup Language DTDs are used by SGML applications, such as HTML, to specify rules for documents of a particular type, including a set of elements and entity declarations. As we stated previously a valid well-formed XML document conforms to a DTD



The Document Type Structure

<!DOCTYPE <root tag>

[<!ELEMENT <name>(<components>)>

. . . more elements . . .

*elements comprise of a pair of tags plus the enclosed data

The description of an element consists of its name (tag), and a parenthesized description of any nested tags. Includes order of subtags and their multiplicity. Leaves (text elements) have #PC-DATA (Parsed Character DATA) in place of nested tags.i.e.

The following is a document type for the previous example

<!DOCTYPE BARS [

<!ELEMENT BARS (BEER*)>

<!ELEMENT BEER (NAME, MANF,SERVEDAT, PRIZE)>



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```
<!ELEMENT NAME (#PCDATA)>
<!ELEMENT MANF (#PCDATA)>
<!ELEMENT SERVEDAT (NAME, ADDR)>
<!ELEMENT NAME (#PCDATA)>
<!ELEMENT ADDR (#PCDATA)>
<!ELEMENT PRIZE (YEAR, AWARD)>
<!ELEMENT YEAR (#PCDATA)>
<!ELEMENT YEAR (#PCDATA)>
```

Using the DTD

1. Set standalone = “no”.
2. Either:
 - Include the DTD as a preamble of the XML document, or



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- Follow DOCTYPE and the <root tag> by SYSTEM and a path to the file where the DTD can be found.

```
<!DOCTYPE BARS [  
  <!ELEMENT BARS (BEER*)>  
  <!ELEMENT BEER (NAME, MANF,SERVEDAT, PRIZE)>  
<!ELEMENT NAME (#PCDATA)>  
  <!ELEMENT MANF (#PCDATA)>  
  <!ELEMENT SERVEDAT (NAME, ADDR)>  
  <!ELEMENT NAME (#PCDATA)>  
  <!ELEMENT ADDR (#PCDATA)>  
  <!ELEMENT PRIZE (YEAR, AWARD)>  
  <!ELEMENT YEAR (#PCDATA)>  
  <!ELEMENT YEAR (#PCDATA)>  
  <?xml version = "1.0" standalone = "no" ?>
```





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```
<BARS>
<BEER><NAME>Bud</NAME>
<MANF>A.B</MANF>
<SERVEDAT><NAME>JOES BAR</NAME>
<ADDR></ADDR>MAPLE</SERVEDAT>
</BEER>
<BEER>><NAME>Bud</NAME>
<MANF>A.B</MANF> <PRIZE>
<YEAR>1995</YEAR><AWARD>GOLD</AWARD>
</PRIZE> </BEER>
...
</BARS>
```

Example . Describe the syntax of a well formed XML document (3 marks).



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Solution: A well-formed document conforms to the XML syntax rules; e.g. if a start-tag (`<` `>`) appears without a corresponding end-tag (`</>`), it is not well-formed. A document not well-formed is not in XML; a conforming parser is disallowed from processing it. Valid. A valid document additionally conforms to semantic rules, either user-defined or in an XML schema, especially DTD; e.g. if a document contains an undefined element, then it is not valid; a validating parser is disallowed from processing it. i.e. `<book>This is a book... </book>`

□



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Learning Activities

Revision Questions

EXERCISE 1. ✍ Define XML stating its significance in web development (2 marks).

EXERCISE 2. ✍ Define Semi Structured Data

EXERCISE 3. ✍ Describe the meaning of the following Terms in XML

1. #PCDATA
2. *
3. <?XML

EXERCISE 4. ✍ Differentiate between XML DTD and XHTML DTD



References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp
- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



Solutions to Exercises

Exercise 1. The Extensible Markup Language (XML) is a general-purpose specification for creating custom markup languages. It is classified as an extensible language, because it allows the user to define the mark-up elements.

Exercise 1

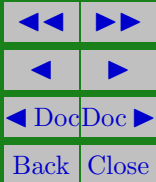


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Exercise 2. Put answer

Exercise 2



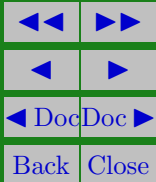


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Exercise 3. Put answer

Exercise 3





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Exercise 4. Put answer

Exercise 4

